

MUHAMMAD SOHAIB SAJID

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PROFILE

Third-year AI student at ITU Lahore with coursework spanning machine learning, MLOps, and reinforcement learning fundamentals, paired with hands-on applied ML experience. Built a machine learning model entirely from scratch (custom SVM, no ML libraries) to validate algorithmic understanding beyond framework usage. Experience translating operational requirements into model and data design during an internship at Pakistan Tobacco Company, working directly between a computer vision initiative and floor operations stakeholders. Comfortable evaluating data quality, model behavior, and tool trade-offs, and communicating findings across technical and non-technical audiences.

EXPERIENCE

Data Annotation & AI Support Intern | Pakistan Tobacco Company (PTC)

Summer 2024

- Worked directly between a computer vision R&D effort and floor operations, translating real-world production requirements into annotation guidelines and validation criteria the model needed to meet.
- Supported adaptation and testing of an anomaly-detection model for real-time conveyor-belt deployment, iterating on data quality based on observed model failure cases.
- Applied validation checks and consistency reviews to training data, directly shaping data design decisions that fed the model pipeline.
- Gained first-hand exposure to translating an ops-floor problem into a research-and-development question, and back into a deployable process.

Full-Stack AI & Mobile Developer | Freelance

Jun 2024 – Present

- Evaluate model and tool choices against real client constraints (data availability, latency, cost, access limits), then design and implement accordingly.
- Built and deployed an AI-driven web application, including evaluating architecture and data-storage trade-offs (B-Trees, hashing) for retrieval performance.
- Migrated a live product across multiple LLM providers (Llama-3, Mistral-7B, Groq) after diagnosing access and quota constraints — a direct exercise in judging when a model or provider signal is reliable versus a temporary limitation.

RESEARCH-RELEVANT PROJECTS

Driver Drowsiness Detection — Built From Scratch

- Implemented a custom SMO-based Support Vector Machine in pure NumPy — no ML libraries — to classify drowsiness states, requiring direct reasoning about the optimization and decision boundary rather than treating the model as a black box.
- Designed the feature pipeline from scratch: Haar cascade face detection with EAR/MAR (eye/mouth aspect ratio) feature extraction, and a 15-frame state machine to judge signal versus noise in frame-level predictions before triggering a detection.

JobLens — AI Career Platform (React, FastAPI, PostgreSQL, LLMs)

- Built skill-gap analysis and CV-to-job matching pipelines requiring careful data design between unstructured resume text and structured job requirements.
- Designed real-time voice mock interviews with behavioral confidence detection, evaluating model output reliability under live conditions.

SKILLS & COMPETENCIES

ML & Research Foundations: Machine learning (linear/logistic regression, SVMs, decision trees, ensemble methods, neural networks), MLOps, reinforcement learning fundamentals (coursework), PyTorch, NumPy

Data Design & Ops Translation: Data annotation & quality validation, feature engineering, requirement translation between operational and technical teams, data pipeline design (PostgreSQL, Kafka, Spark, FAISS)

Engineering (supporting): Python, JavaScript, React, FastAPI, SQL, MongoDB, Git & GitHub

EDUCATION

BS Artificial Intelligence | Information Technology University (ITU), Lahore

Expected 2028

CGPA: 3.32 / 4.0

Relevant coursework: Artificial Intelligence, Machine Learning, Programming for AI, MLOps (incl. reinforcement learning fundamentals)